

Engineering Germany's Industrial Success

Engineering Societies, Patents, and the 19th-Century Catch-Up

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Few economies have ever reached the global technological frontier, and fewer still have stayed there. Existing work has identified formal inputs to that transition — human capital, patent rights, university research — but the organisational layer through which these combine into innovative effort has received less attention. We isolate one such layer. Germany's late-19th-century catch-up to global leadership in chemistry, electrical machinery, and precision instruments was coordinated in part by the German Engineering Society, which between 1856 and 1925 built a national network of local chapters organising practical engineering knowledge through monthly meetings, factory visits, and a weekly technical journal. Exploiting the staggered founding of 22 chapters over 1877–1918 in a Callaway–Sant'Anna design on the Streb–Baten high-value patent corpus, we estimate that chapter founding raises annual patenting by approximately four percent, with flat pre-trends and a null on the patent classes outside the Society's scope. The response operates through amplifying existing capacity, and through practical owner-engineers rather than credentialed graduates. A chapter-proximate cross-sectional specification documents a manufacturing-GDP premium for counties near pre-1918 chapter headquarters through the mid-twentieth century. The professional organisation of practical knowledge emerges as a distinct channel for innovation capacity, complementary to formal education and patent rights.

1. Introduction

Why national economies differ so persistently in their innovative capacity is among the defining questions of modern growth economics. Growth theory pinpoints what sustained innovation implies for development (Romer, 1990; Aghion and Howitt, 1992) but says less about where the capacity to innovate comes from in the historical record. An empirical literature has begun to fill in particular formal inputs to that capacity. Squicciarini and Voigtländer (2015) link upper-tail human capital, measured by subscriptions to the *Encyclopédie*, to French city growth during industrialisation. Moser (2005) shows that patent institutions shape the direction of innovation more than its volume. Dittmar and Meisenzahl (2022) document the effect of research-university foundings on innovation in 19th-century Germany. Donges, Meier, and Silva (2023) identify the effect of Napoleonic-era legal-institutional reforms in the same setting. Across this work the focus has been on credentialed human capital, formal rules, and university science. What we know less about is how these formal inputs were combined on the ground — through what organisational forms the credentialed knowledge of the academy and the practical knowledge of the factory floor came to meet.

The case for organisation as a separable input is both theoretical and empirical. Kelly, Mokyr, and Ó Gráda (2023) argue that Britain’s industrial revolution rested on a substantial stock of artisan mechanical skill: tacit, problem-specific knowledge accumulated outside the academy by shop-floor practitioners. Mokyr (2002) characterises industrialisation more broadly as an *Industrial Enlightenment* in which propositional and prescriptive knowledge diffused across a porous boundary between savants and artisans; the diffusion mechanism, on his account, is the institutional fabric through which the two communities communicated. The most closely related empirical work is Cinnirella, Hornung, and Koschnick (2025), who show that 18th- and early-19th-century German economic societies — characterised by membership rolls, regular meetings, and the publication of useful knowledge — predict subsequent patenting in their host regions. What is missing in the existing evidence is a causal estimate of the organisational layer in the specific case of *technical-professional* knowledge in the 19th century — the period and the substance through which the second industrial revolution actually proceeded. In Germany during 1856–1925, that organisational layer took a particular form: the local chapter of the national engineering society. Whether those chapters caused engineering activity, or merely tracked it, is the question we take up.

The German case is unusually well suited to the question for three reasons. First, by 1900 Germany had overtaken Britain in high-value patenting per capita and led global markets in

1. Introduction

chemistry, electrical machinery, and precision instruments — the canonical late-19th-century catch-up against which any account of innovation capacity must be measured. Second, political and patent unification in 1871 and 1877 produced a single national framework whose variation is local: the staggered founding of 22 local chapters over 1877–1918 provides within-country, within-period variation that cross-country and pure cross-sectional designs cannot replicate. Third, the counterpart literatures impose binding constraints on what existing evidence can identify. The cross-country evidence in Maloney and Valencia Caicedo (2022) — engineer density predicts subsequent income, identified through the 1862 US Morrill Act — cannot speak to organisational forms that combine engineers into innovative activity within a given national supply of engineering labour. The patent-inventor-level evidence in Hanlon (2025), documenting the rise of the credentialed engineer in 19th-century Britain, cannot identify how engineers were coordinated in a country in which professional organisation took a markedly different form. We complement both, holding national patent law, national education policy, and the broad supply of engineers approximately constant, and identifying the effect of a particular organisational form on patent activity inside the country and within the period in which engineers were becoming a recognised professional category.

The empirical strategy uses the staggered chapter rollout to identify the causal effect of chapter founding on high-value patenting. The patent outcome is drawn from the Streb, Baten, and Yin (2006) corpus of patents filed at the Imperial Patent Office between 1877 and 1918 and renewed beyond ten years — a selection that screens for commercial value, since cumulative renewal fees through the tenth year amounted to roughly two-and-a-half years of industrial wages (Bry, 1960). We assign each patent to a 1900 statistical unit via the inventor’s residence address, assign each statistical unit to a chapter by majority-area overlay on the chapter-territory shapefile, and compare patenting in statistical units about to be covered by a newly-founded chapter against patenting in not-yet-covered units. Because chapter effects are heterogeneous across founding cohorts and dynamic within the post-period — chapters took time to recruit members, organise meeting calendars, and embed in the local industrial fabric — standard two-way fixed effects estimates attenuate substantially. We use the doubly-robust Callaway–Sant’Anna estimator (Callaway and Sant’Anna, 2021) as the workhorse and corroborate with the de Chaisemartin–D’Haultfoeuille switchers-vs-stayers estimator (Chaisemartin and D’Haultfoeuille, 2024), which rests on a different identifying assumption.

Chapter founding raises annual high-value patenting in the canonical sample by approximately 4.2 percent on average over the post-treatment window. The pre-trend block is flat across the

1. Introduction

15-year pre-period and the joint test does not reject parallel trends. The post-treatment response builds gradually, from approximately zero at the founding year to roughly nine percent fifteen years later, consistent with a process in which chapters take time to recruit members, embed in the local industrial fabric, and begin coordinating practical knowledge through their meeting calendar. Decomposing the patent corpus along the Society’s own 1925 committee structure delivers what we view as the design’s most direct identification check. On the 49 patent classes the Society organised committees around — chemistry, machinery, electrical engineering, transport, metallurgy, mechanised textiles — the simple ATT is statistically distinguishable from zero and quantitatively close to the headline. On the 40 patent classes for which the Society had no committee — windmills, watermills, mechanical clocks, traditional crafts — the simple ATT is statistically indistinguishable from zero. The bucket boundary is defined *ex ante* from the 1925 committee structure, fixed before any of the outcomes we observe, not from an *ex post* search for the cleanest contrast.

The result survives a substantial robustness battery. Across fourteen specifications — alternative heterogeneity-robust estimators (Sun and Abraham, 2021; Borusyak, Jaravel, and Spiess, 2024; Arkhangelsky et al., 2021; Chaisemartin and D’Haultfœuille, 2024), sample restrictions, specification variants including PPML on raw counts, two donut-hole designs, leave-one-chapter-out estimation, and a 200-replication permutation test — the CS-DiD-family point estimates lie between 2.0 and 7.0 percent per year, and PPML implies roughly 15 percent on counts. Across the doubly-robust estimators that incorporate inverse-propensity reweighting or pre-period matching — CS-DiD, dCDH, and the Borusyak-Jaravel-Spiess imputation — the estimates cluster tightly between 4.2 and 4.8 percent. Two donut-hole specifications — a temporal donut that drops the years immediately around founding, and a spatial donut that compares statistical units 10–50 kilometres from a chapter against units 50–100 kilometres away — both return point estimates close to the full-sample headline. The same spatial donut applied to the founding of engineering schools returns flat noise: the chapter effect is not a downstream consequence of the educational shock that produced the chapter’s members.

Three pieces of within-design evidence pin down the mechanism. First, holding the chapter constant, statistical units within a chapter’s territory with denser 1900 membership produced more post-founding patenting — the partial effect is 0.68 percent per percent of member density after chapter fixed effects, 1871 population, and railway controls are absorbed. Members are themselves the patenters, not just markers of an industrial region. Second, regressing each chapter’s post-founding sectoral patent share on its pre-founding sectoral share, bucket by bucket, yields a slope that approaches unity on the bucket of pre-existing specialty and falls

1. Introduction

to zero on the pre-industrial placebo. Chapters reinforced what was already there; they did not impose a uniform technological program. Third, an individual linear probability model on the matched membership–patent registry shows that owner-practitioners — the members Gispén (1989) describes as the carriers of “shop culture,” trained on the factory floor rather than in the technical universities — were 3.8 percentage points more likely to appear as named patent inventors than salaried employee engineers, while members holding the Dipl.-Ing. credential of the technical universities were modestly *less* likely. The patenting response operated through the engineers closest to the shop floor, not through credentialed graduates. Together the three pieces of evidence rule out, respectively, the alternatives that chapters were regional markers without a within-region effect, that they imported new technology into passive regions, and that they operated through academic-engineering spillovers.

The patent-level estimates establish a causal effect within the panel window 1877–1918. Whether that effect translated into long-run economic outcomes is a harder question, and one for which we can offer only suggestive evidence. The long-run output data (Peters, 2022) cover West Germany only and begin in 1933; the within-country staggered design is not directly available. We use a spatial donut-hole specification instead, comparing modern counties 10–50 kilometres from a pre-1918 chapter headquarters against counties 50–100 kilometres away, dropping the host city and counties beyond 100 kilometres. The manufacturing-GDP premium in the chapter-proximate ring is 24–27 log points (roughly 27–31 percent) through 1935–1966 and fades on aggregate GDP per worker to roughly four percent by the present. Two features of the long-run pattern align with the patent-level mechanism: the premium is concentrated on manufacturing, where the patent-level effect operated, and is much weaker on aggregate output; and it survives both world wars, suggesting persistent regional human capital and organisational know-how rather than persistent physical infrastructure. We do not claim causal identification for the long-run results; we report them as evidence consistent with the patent-level mechanism.

The paper contributes to three related literatures. The most direct engagement is with the literature on *engineers and growth*. The cross-country evidence in Maloney and Valencia Caicedo (2022) establishes that engineer density predicts subsequent income through the 1862 US Morrill Act; Hanlon (2025) traces the patent-level rise of the credentialed engineer in 19th-century Britain. Our contribution is within-country, period-specific, and organisational: with patent law, education policy, and the aggregate supply of engineers fixed at the national level, the design isolates what the local organisation of engineers added to patent activity at the moment the profession was taking shape.

2. Historical Setting and Data

A second engagement is with the literature on *the German innovation system in the long 19th century*. Cinnirella and Streb (2017) document the role of literacy and human capital in post-Malthusian Prussia; Donges, Meier, and Silva (2023) use the staggered Napoleonic introduction of legal-institutional reforms to identify their effect on innovation; Dittmar and Meisenzahl (2022) establish the effect of research-university foundations. We add a non-university, non-formal-institutional channel to this set, identified using within-country staggered variation comparable in design to Donges, Meier, and Silva but operating on a different treatment.

A third engagement is with the literature on *knowledge networks and economic societies*. Cinnirella, Hornung, and Koschnick (2025) establish that 18th- and early-19th-century German economic societies predict subsequent patenting in their host regions; we extend the question to the 19th-century technical-professional case. The Society shares the organisational form of these earlier societies, but we focus on societies with members that were engineers attending factory visits and machinery demonstrations; and our staggered timing variation delivers a causal estimate the cross-sectional design in their work cannot. The two studies are complementary: Cinnirella, Hornung, and Koschnick establish that knowledge-network organisation matters for innovation; we identify the channel through which it operated in the second industrial revolution and document the within-design placebo that pins down the mechanism for the case of engineers.

Outline. Section 2 describes the historical setting and the data. Section 3 reports the cross-sectional facts that motivate the causal design. Section 4 sets up the empirical strategy and the bias of two-way fixed effects under staggered adoption. Section 5 presents the headline result, robustness, and the donut-hole identification check. Section 6 documents the mechanism. Section 7 presents the long-run manufacturing premium. Section 8 discusses the results and Section 9 concludes.

2. Historical Setting and Data

2.1. The catch-up puzzle and the engineering society

By 1850 Germany was a politically fragmented agrarian economy decades behind Britain in industrial output. By 1900 it had overtaken Britain in high-value patenting per capita

2. Historical Setting and Data

and led global markets in chemistry, electrical machinery, and precision instruments. The institutional framework conditioning this transition came together in three steps: polytechnic schools were established from 1825 onward (eleven institutions by 1880, upgraded to technical universities by 1899); state engineering schools followed from the 1880s, with 29 institutions by 1900 providing practical training to most working engineers (Gispen, 1989); and political unification (1871) together with a unified patent law (1877, modelled on the British system) created the national innovation system. What this framework did not provide — and what the German Engineering Society filled — was a coordination layer between the academy and the factory floor.

The German Engineering Society (*Verein Deutscher Ingenieure*, VDI) was founded in Berlin in May 1856 by twenty-three young engineers. It operated through a weekly technical journal distributed to every member across the Empire, and through a network of local chapters that organised monthly technical lectures, factory visits, and machinery demonstrations. Chapter leadership was dominated not by university professors but by factory owners and senior practitioners (Gispen, 1989); the society was, in practice, a coordination device for industrial as much as for academic knowledge. Membership grew from 172 in 1856 to more than 27,000 in 1925; chapter coverage expanded from 5 to 50. Figure 1 maps the chapter-headquarters network on 1925 administrative boundaries. The VDI also influenced national policy through the 1877 patent law and through its national standards committee (Gispen, 1989), but the variation we exploit is the local rollout, not the national footprint.

2.2. Data

The patent outcome is drawn from the Streb, Baten, and Yin (2006) corpus of patents filed at the Imperial Patent Office between 1877 and 1918 and renewed beyond ten years ($N = 39,360$ of approximately 380,000 filings). The renewal hurdle is meaningful because maintenance fees were substantial: cumulative fees through year ten amounted to 2,300 Marks, roughly two-and-a-half years of industrial wages (Bry, 1960). Patents surviving the year-ten renewal therefore signal commercial value rather than mere filing activity. The corpus is organised into 89 technology classes. The classification records the application domain of the invention rather than the inventor’s professional discipline: a chemical process applied to textile dyeing, for instance, is recorded under textiles, not chemistry. We aggregate the 89 classes into two principal buckets defined ex ante by the Society’s 1925 technical-committee structure. The first comprises the 49 classes that fall under second-industrial-revolution technologies

2. Historical Setting and Data

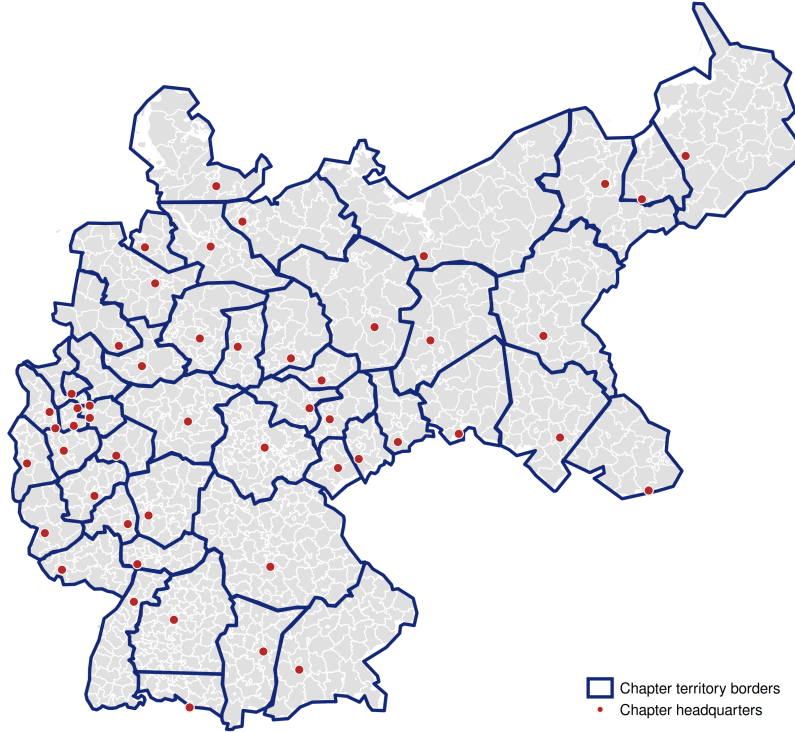


Figure 1: Local chapter headquarters of the German Engineering Society, with 1925 territorial boundaries overlaid on 1900 statistical-unit borders.

(chemistry, machinery, electrical engineering, transport, metallurgy, mechanised textiles) and accounts for roughly three-quarters of the corpus. The second comprises 40 classes spanning windmills, watermills, mechanical clocks, traditional crafts, and small consumer goods, for which the Society had no technical committee; this bucket serves as the placebo outcome throughout. The chemists split from the VDI in 1887 to form a separate society and the electrical engineers split in 1893, biasing our estimates on those sectors downward through the post-split panel years.

We use the 1900 and 1925 VDI membership directories (15,513 and 27,176 hand-transcribed individuals; 26,869 unique individuals after within-name-and-city deduplication across the two directories), each listed with name, professional title, employer, and city of residence. For the engineer-level analysis in Section 6 we link the directory to the Streb–Baten corpus by name using the `reclink2` probabilistic record-linkage routine (Wasi and Flaaen, 2015), producing 644 person-level links. The analysis uses 1900 statistical units as the primary geographic resolution. The German Empire contained 1,020 such units; 493 enter the canonical difference-

3. Stylised Facts

in-differences sample once always-treated units (covered by chapters founded by 1877) and units outside any chapter territory are excluded. Chapter territories are taken from the 1925 yearbook and assigned to statistical units via majority-area shapefile overlay; patents are geocoded via the inventor’s residence address. The eastern territories (East and West Prussia, Posen, parts of Silesia) were predominantly rural and structurally less suited to a technical-professional treatment; the headline strengthens marginally when they are excluded (Appendix C). Historical population comes from Rösler (2023); the 1871 cross-section forms the basis for pre-treatment density and an interaction with year provides a time-varying trend control. The railway network is mapped at the unit-year level from Cederman, Pengl, and Girardin (2025). Long-run outcome data come from Peters (2022) for the 1933–1966 period at the modern-county level (manufacturing GDP, manufacturing employment, GDP per worker) and from the INKAR regional indicators (BBSR, 2019) for 2019. Coverage is West Germany throughout the long-run exercises, since post-1945 division and central planning in the East leave no comparable output series.

2.3. Spatial co-distribution and summary statistics

A first look at the data establishes the cross-sectional fact that motivates the causal design. Figure 2 maps engineers and high-value patents at the statistical-unit level. The spatial co-distribution is tight: regions with more engineers also produced more patents. This is the regularity that any causal claim must navigate around, since chapters were founded where engineers already clustered. The remainder of the paper uses staggered timing within region to break the identification problem this co-distribution creates.

Table 1 reports summary statistics: Panel A covers the full Kaiserreich cross-section of 1,020 statistical units, Panel B the chapter network. The canonical difference-in-differences sample of Section 4 comprises 493 of these units in an annual panel from 1877 to 1918 ($N = 20,706$ unit-year observations); within this sample the patent outcome averages 0.30 high-value patents per unit-year, with a right skew that motivates the $\ln(1 + x)$ functional form used throughout; PPML on raw counts is reported in Appendix C.

3. Stylised Facts

Two cross-sectional facts motivate the causal design. Both establish that there is something to identify; neither delivers identification.

3. Stylised Facts

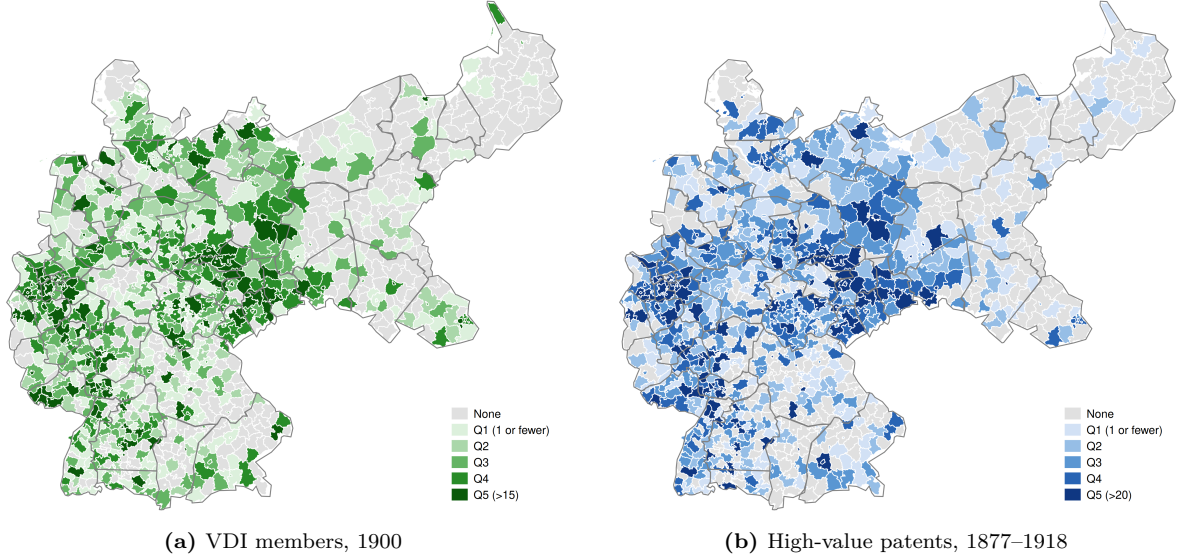


Figure 2: Spatial co-distribution of engineers and high-value patenting at the statistical-unit level. Quintile choropleths on the 1900 boundary file.

Table 1: Summary statistics: 1900 statistical units (Panel A, full Kaiserreich) and local chapters (Panel B).

	Mean	Median	SD	Min	Max
<i>Panel A: All 1900 Statistical Units (full Kaiserreich, $N = 1020$)</i>					
Total high-value patents (1877–1918)	24.9	2	196.1	0	5,668
Engineering	8.5	0	91.1	0	2,802
Chemical	5.9	0	49.2	0	842
Mechanical engineering	3.6	0	22.1	0	581
Patents by Individuals	12.4	1	86.7	0	2,507
Patents by Firms	12.4	0	113.3	0	3,161
Engineering Society members (1900)	13.2	1	61.5	0	1,478
Population, 1871 (000s)	28.7	24	44.6	0	1,029
Railroad km (1885)	32.4	28	27.5	0	226
<i>Panel B: Local Chapters — All Bezirksvereine ($N = 50$)</i>					
Founding year	1882	1878	19.6	1856	1924
Members (1900)	297	259	266	61	1,612

The first is the within-country counterpart to the cross-country regularity that Maloney and Valencia Caicedo (2022) document. Table 2 reports the cross-sectional OLS of $\ln(1 + \text{patents } 1877\text{--}1918)$ on $\ln(1 + \text{engineers } 1900)$ at two administrative resolutions: modern municipality ($N = 10,736$) and modern county ($N = 315$). The coefficient lies between 0.49 and

3. Stylised Facts

0.71 across columns, with state fixed effects, 1871 population, and pre-treatment geographic controls. A ten-percent increase in 1900 engineer density predicts a five- to seven-percent increase in high-value patenting over the next four decades. The association is stable across resolutions, robust to dropping the top one percent of units by patent count, and quantitatively comparable to the engineer-share-and-growth relationship that Maloney and Valencia Caicedo (2022) document on cross-country data.

Table 2: Cross-sectional OLS: 1900 engineer density and high-value patenting, 1877–1918.

	Municipality (N = 10,736)		District (N = 315)	
	(1)	(2)	(3)	(4)
	All	Engineering	All	Engineering
ln(1+Engineers 1900)	0.709*** (0.023)	0.488*** (0.023)	0.636*** (0.047)	0.588*** (0.044)
ln(population 1871)	0.049*** (0.008)	-0.000 (0.005)	0.496*** (0.136)	0.415*** (0.118)
<i>Controls:</i>				
Geography	✓	✓	✓	✓
Soil & climate	✓	✓	✓	✓
State FE	✓	✓	✓	✓
R ²	0.606	0.582	0.657	0.626

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The second fact is that the cross-sectional pattern persists into long-run economic outcomes. Regressing West-German manufacturing GDP (Peters, 2022) on 1900 engineer density at the modern-county level yields coefficients near 0.30 in 1935 and 0.21 in 1966 — a persistence of three generations through both world wars. The analogous coefficient on aggregate GDP per worker is 0.14 in 1935 and fades to roughly 0.03 by 2019. Manufacturing responds several times more strongly than aggregate output, and the manufacturing effect survives both wars. Appendix Table 10 reports the underlying regressions. Section 7 returns to this pattern with a design-based counterpart.

These two facts establish a robust regularity but do not identify a causal effect. Chapters were founded in places that were already economically important, with stronger industrial fabric and denser population. The cross-sectional coefficient could reflect a causal effect of

4. Empirical Strategy

organisation on patenting, a selection effect from chapter placement, or both. The remainder of the paper uses within-region variation in the timing of chapter founding to separate them.

4. Empirical Strategy

4.1. Staggered chapter rollout

Twenty-two chapters were founded between 1877 and 1918. The earliest switchers in our identification window were Schleswig-Holstein and East Prussia (both 1880); the latest was Mosel (1912). Figure 3 plots the resulting identification map. Each polygon is a chapter territory, labelled by founding year. Grey territories belong to the chapters that contribute no within-panel timing variation: the 25 chapters already founded when the patent panel opens in 1877, treated throughout the window and dropped from the treated set, and the three chapters founded after 1918 — Elbing and Lübeck (1921) and Osnabrück (1924) — whose statistical units serve as never-treated controls. Orange territories are the 22 switching chapters founded between 1877 and 1918 whose staggered timing drives identification. The orange-shaded territories span the full geographic extent of the Empire. Identification in what follows comes from comparing patenting in statistical units within an orange territory about to be covered by a newly-founded chapter against patenting in units not yet covered. A chart of switching dates is provided in Appendix C (Figure 7).

4.2. Specification and identification assumptions

The baseline specification is

$$\ln(1 + \text{Patents})_{it} = \beta D_{it} + \gamma' \mathbf{X}_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where i indexes statistical units, t indexes years from 1877 to 1918, and $D_{it} = 1$ once the chapter covering unit i has been founded. The control vector $\mathbf{X}_{it}$ contains time-varying railway kilometres and a pre-treatment population trend (1871 population $\times$ year); α_i and δ_t are statistical-unit and calendar-year fixed effects. The panel comprises $N = 20,706$ unit-years across 493 statistical units. Identification requires no anticipation, parallel trends conditional on $\mathbf{X}$, and no spillover across territorial boundaries. The first two assumptions are testable

4. Empirical Strategy

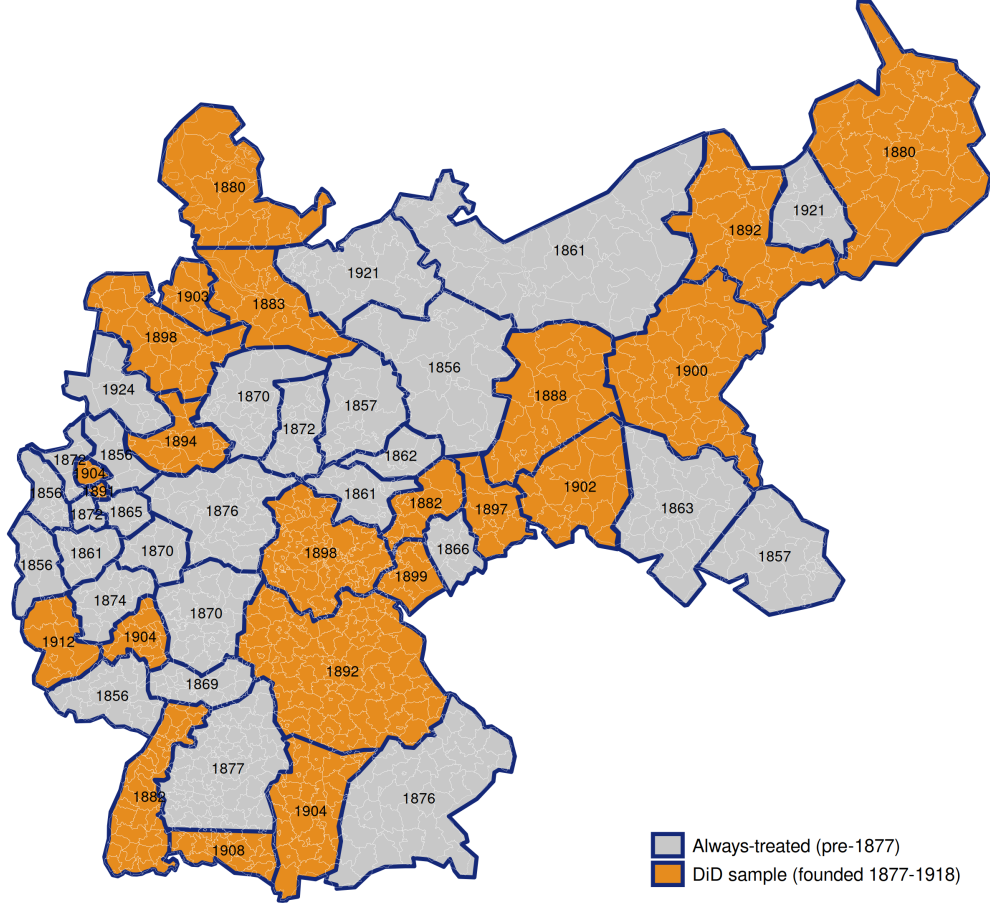


Figure 3: The identification map. Chapter territories on the 1900 statistical-unit boundary file, labelled by chapter founding year. Grey: non-switching territories — the 25 chapters founded by the 1877 start of the patent panel (always-treated, dropped from the treated set) and the three chapters founded after 1918 (never-treated). Orange: the 22 switching chapters founded over 1877–1918 whose staggered timing identifies the chapter effect.

through the event-study aggregation; the third is examined through a donut-hole design in Section 5.

4.3. Two-way fixed effects under staggered adoption

Standard two-way fixed effects is biased under staggered adoption with heterogeneous and dynamic treatment effects. The Goodman-Bacon (2021) decomposition shows that TWFE is a weighted average of all possible 2×2 comparisons between treated and control unit-years, including “forbidden” comparisons in which already-treated cohorts serve as controls; when

4. Empirical Strategy

treatment effects build up post-founding the forbidden comparisons enter with negative weight and attenuate the TWFE coefficient. On our canonical sample the TWFE coefficient on D_{it} is 0.014 — roughly a factor of three below the CS-DiD estimate of 0.042 we report below. The Goodman-Bacon decomposition (Appendix B) shows that 99.8 percent of TWFE weight on this sample comes from legitimate timing-group comparisons, since we drop always-treated chapters by construction, so the forbidden-comparison block is empirically negligible. The TWFE–CS-DiD gap on our sample is therefore not a forbidden-comparison artefact; it is the absence of doubly-robust inverse-propensity reweighting in TWFE, which pools heterogeneous treatment dynamics across cohorts without rebalancing on covariates.

4.4. Callaway–Sant’Anna and a corroborating estimator

We use the Callaway and Sant’Anna (2021) estimator as the workhorse. The estimator builds cohort-time-specific average treatment effects

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid G_i = g, t \geq g], \quad (2)$$

where g indexes a chapter’s founding year and t indexes calendar time, then aggregates across (g, t) pairs after estimation. We implement the doubly-robust inverse-propensity-weighted variant using not-yet-treated units as controls. The estimator is consistent if either the propensity score for being in cohort g conditional on $\mathbf{X}$ is correctly specified or if the outcome regression on never-treated controls is correctly specified. Three aggregations are reported: the simple ATT, the cohort-size-weighted post-treatment average, and the event-study aggregation. Standard errors come from the multiplier bootstrap with 999 replications.

The CS-DiD estimator rests on conditional parallel trends, which is testable in the pre-period but untestable in the post-period. We bring in the Chaisemartin and D’Haultfoeuille (2024) switchers-vs-stayers estimator (dCDH) as a corroborating estimator on a different identifying assumption. dCDH compares treatment-status switchers to stayers at each panel year and does not apply inverse-propensity reweighting. On the canonical sample the dCDH simple ATT is 0.048 (SE 0.005) with a flat pre-period (coefficient -0.0003) and a joint placebo test that does not reject.

5. Results

5.1. Headline result and within-design placebo

The central result is reported in Figure 4 and Table 3. Chapter founding raises annual high-value patenting in the canonical 22-switcher sample by approximately 4.2 percent on average over the post-treatment window (simple ATT = 0.042, multiplier-bootstrap SE 0.014, $p = 0.003$). The pre-trend block is flat: none of the 15 pre-period coefficients is significant at the five-percent level, the average pre-period effect is essentially zero, and the joint test of parallel trends does not reject. The post-treatment block rises monotonically from approximately zero at $t = 0$ (the founding year), to roughly three percent at $t = +5$, to six to eight percent at $t = +10$, to nine to ten percent at $t = +15$. The gradual build-up matches a simple recruitment-and-meeting story: chapters did not switch on at full strength; they grew their membership and meeting calendar and gradually became embedded in the local industrial fabric.¹

The decomposition along the Society’s 1925 committee structure delivers the within-design placebo. On the 49 patent classes the Society organised committees around — chemistry, machinery, electrical engineering, transport, metallurgy, mechanised textiles — the simple ATT is 0.037 (SE 0.011), close to the headline. On the 40 patent classes the Society had no committee for — windmills, watermills, mechanical clocks, traditional crafts — the simple ATT is 0.016 (SE 0.010), and we cannot reject the null at conventional levels. Where the Society organised committees, patenting accelerated; where it did not, it did not. The bucket boundary is drawn from the 1925 yearbook, fixed before any of the patent outcomes we observe; the contrast is therefore an ex ante prediction the data could have rejected and did not.²

A subsample restriction to peacetime years (1877–1913) yields a simple ATT identical to the full window; the result is not driven by wartime patent-office disruption.

¹The Streb–Baten corpus dates patents by grant year, not filing year. Under the 1877 Imperial Patent Act, substantive examination (*Vorprüfung*) and the mandatory eight-week public opposition period typically introduced a lag of one to two years between filing and grant: a patent applied for in year g would not appear in our data until $g + 1$ or $g + 2$. Part of the observed $t = +1$ to $t = +2$ ramp therefore reflects this mechanical dating lag rather than treatment-effect dynamics; the recruitment-and-meeting interpretation remains the leading reading of the slope at longer horizons.

²The pre-industrial bucket coefficient is positive but statistically indistinguishable from zero ($t \approx 1.5$); a residual general-equilibrium effect of the post-1880 industrial boom on all patenting cannot be ruled out, but the magnitude is well below the modern bucket and the boundary was drawn ex ante from the 1925 committee structure.

5. Results

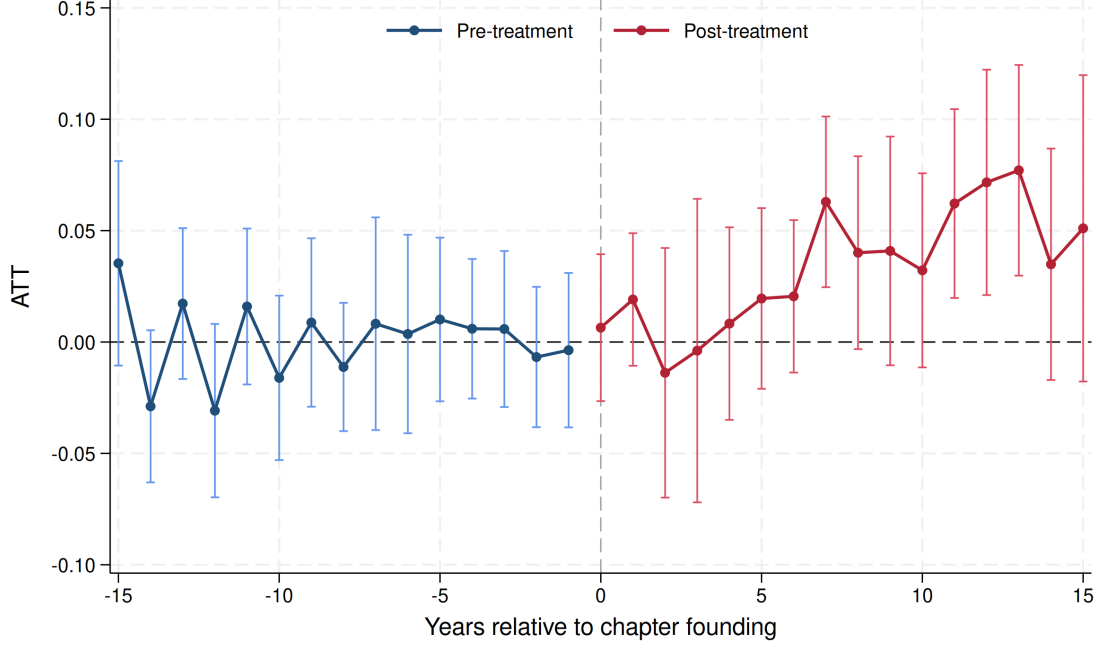


Figure 4: Callaway-Sant’Anna event study on $\ln(1+\text{patents})_{it}$, ± 15 event-time window, canonical 22-switcher sample. Vertical bars: 95% confidence intervals from the multiplier bootstrap with 999 replications.

Table 3: Callaway-Sant’Anna headline ATT: total high-value patents and decomposition into second-industrial-revolution technologies (49 classes) and pre-industrial / artisanal technologies (40 classes; placebo). Multiplier-bootstrap standard errors in parentheses. Sample: 22 switching chapters, 1877–1918, $N = 20,706$ unit-years across 493 statistical units. Method: Callaway-Sant’Anna DiD (`dripw`); controls: rail + pop $\times$ year.

	Total (all 89 cl.)	2nd Ind. Rev. (49 cl.)	Pre-Industrial (40 cl., placebo)
ATT (CS-DiD, ± 15)	0.0419*** (0.0141)	0.0366*** (0.0114)	0.0157 (0.0104)
<i>Implied effect</i>	+4.2%/yr	+3.7%/yr	+1.6%
Sample: 22 switching chapters, 493 stat units, 1877–1918 (20,706 obs)			
Method: Callaway-Sant’Anna DiD (<code>dripw</code>); controls: rail + pop $\times$ year			

5.2. Robustness

Table 4 reports fourteen specifications grouped into three categories: alternative heterogeneity-robust estimators, sample restrictions, and specification-and-inference adjustments. The CS-

5. Results

DiD–family point estimates lie between 0.020 and 0.070 across the table; PPML on raw counts implies roughly 15 percent on counts. The permutation-test p -value of 0.12 (two-sided) and the synthetic-DiD insignificance are the two rows that fall outside conventional significance thresholds; we discuss both below.

Table 4: Robustness battery for the headline CS-DiD coefficient.

Specification	$\hat{\beta}$	SE	Note
Headline: CS-DiD (Annual, ± 15)	0.042 ^{***}	(0.014)	22 chapters, 1877–1918
<i>Group A: Alternative heterogeneity-robust estimators</i>			
dCDH (Chaisemartin and D’Haultfœuille, 2024)	0.048 ^{***}	(0.005)	Pre-trend -0.0003 , n.s.
Sun and Abraham (2021)	0.020 ^{***}	(0.004)	SA pre-trend 0.013 ^{***} , flagged
Borusyak, Jaravel, and Spiess (2024)	0.043 ^{***}	(0.004)	Imputation estimator
Synthetic DiD (Arkhangelsky et al., 2021)	0.027	(0.019)	Direction confirmed; n.s.
<i>Group B: Sample restrictions</i>			
CS-DiD, exclude eastern territories	0.049 ^{***}	(0.018)	13 cohorts retained
CS-DiD, pre-WWI sample (1877–1913)	0.042 ^{***}	(0.014)	Drops wartime years
CS-DiD, balanced panel (≥ 10 pre-yrs)	0.070 ^{***}	(0.016)	Drops 3 short-pre chapters
<i>Group C: Specification and inference</i>			
CS-DiD, annual population control	0.043 ^{**}	(0.018)	Bad-control test
Temporal donut (± 10 yrs around founding)	0.047 ^{**}	(0.024)	Stat unit
Spatial donut (controls 50–100 km from HQ)	0.059 [*]	(0.035)	Host-city confound removed
PPML count regression (no ln transform)	0.140 [*]	(0.081)	$\sim 15\%$ on counts
Leave-one-chapter-out (range across drops)	[0.015, 0.047]	—	20 of 22 drops sig. at 5%
Placebo permutation (200 shuffles)	0.042	$p = 0.12$	Two-sided; one-sided ≈ 0.06
TWFE–Conley (200 km, 5y HAC) (Conley, 1999)	0.013 [*]	(0.008)	TWFE estimate

Notes. Across the CS-DiD–family rows $\hat{\beta} \in [0.020, 0.070]$; PPML implies $\sim 15\%$ on counts. Sun–Abraham is flagged because its pre-trend (0.013^{***}) violates parallel trends; CS-DiD, dCDH, and BJS show flat pre-trends. Significance: ^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

Four patterns in the robustness table are worth noting. First, the doubly-robust estimators that incorporate inverse-propensity reweighting or pre-period matching — CS-DiD, dCDH, and the Borusyak–Jaravel–Spiess imputation — cluster tightly between 0.042 and 0.048; the synthetic-DiD estimate of 0.027 is lower because the pre-trend-match routine trims chapters whose pre-period trajectory the synthetic-control routine cannot match within tolerance. The cross-estimator pattern is consistent with a true underlying effect of approximately four to five percent per year on covariate-balanced comparisons. Second, the Sun–Abraham interaction-weighted estimator returns a smaller point estimate of 0.020 but with a pre-trend coefficient of 0.013 that is itself statistically significant at the one-percent level. The pre-trend violation is estimator-specific — CS-DiD, dCDH, and BJS all return flat pre-trends on the same sample

5. Results

— so we read it as an SA-internal sensitivity rather than an identification failure. Third, the leave-one-chapter-out range is bounded below by the Mosel drop (founded 1912, only six post-treatment years); 20 of 22 single-chapter drops are significant at the five-percent level. The headline is not driven by any single chapter. Fourth, the permutation p -value reflects the resolution limit of randomisation inference with only 22 switching chapters: reassigning founding years across so few units produces a coarse reference distribution, and the one-sided p -value of approximately 0.06 sits at the margin of conventional significance. We read the permutation evidence as consistent with, rather than contradicting, the multiplier-bootstrap inference.

5.3. Chapters concentrate innovation; engineering schools do not

A concern with the headline is that the chapter effect could be a downstream consequence of the educational system that produced its members. Twenty-nine state engineering schools were founded between the 1880s and the 1900s, with substantial geographic overlap with the chapter network. If the patenting response we attribute to chapters were actually driven by engineering-school spillovers — and chapters were merely the visible marker of an underlying educational shock — then a placebo using engineering-school founding instead of chapter founding should return a similar pattern. It does not. Figure 5 reports two parallel donut-hole event studies. The left panel uses chapter founding as the treatment; the right panel uses engineering-school founding. In both cases the treatment ring is statistical units 10–50 km from the founding event, the control ring is 50–100 km, and the host city (< 10 km) is dropped. The chapter ring effect rises monotonically over event time; the engineering-school ring effect is statistically indistinguishable from zero across the same event-time window.

The interpretation is straightforward. Engineering schools train engineers who, on graduation, disperse to wherever industrial employment is available; the local-treatment-ring effect on patents in the host region need not be positive. Chapters coordinate engineers who are already located in the region; the local-treatment-ring effect should rise as the chapter’s roster grows. The figure delivers exactly that pattern, ruling out the educational-spillover alternative as the source of the headline.

6. Mechanism: Amplifying Existing Capacity

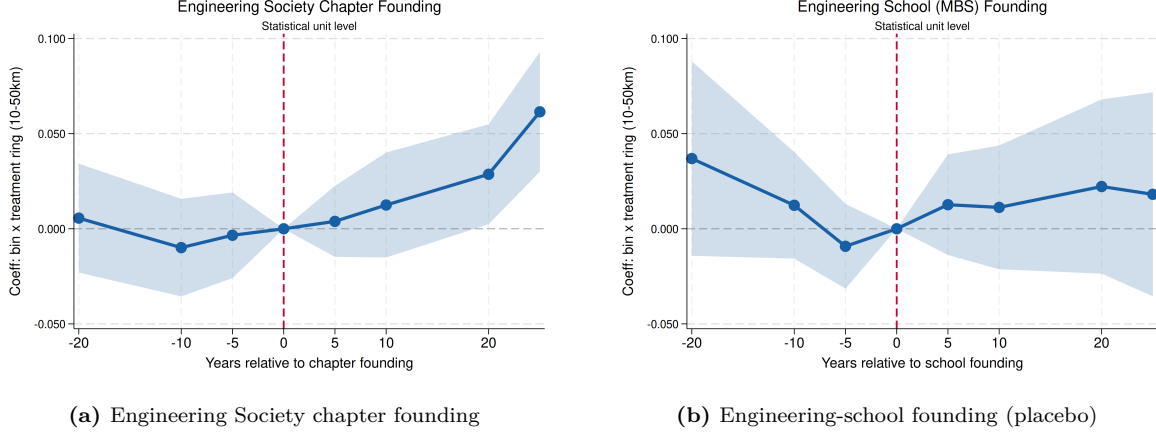


Figure 5: Donut-hole event studies on the patent outcome. Treatment ring: 10–50 km from the founding event. Control ring: 50–100 km. Host city (< 10 km) dropped.

6. Mechanism: Amplifying Existing Capacity

The headline establishes that chapter founding raised patenting. It does not tell us how. Two competing mechanisms are consistent with the headline. Under a creation reading, chapters brought new technology into regions that previously lacked it: the patenting response is concentrated in newly-introduced sectors. Under an amplification reading, chapters strengthened engineering activity that was already incipient in the region: the response is concentrated in sectors that already had a foothold. The two readings differ in their long-run interpretation, in their policy implications, and in their relationship to the cross-country regularity that Maloney and Valencia Caicedo (2022) document. Three pieces of within-design evidence point consistently to the amplification reading.

6.1. Member density predicts patenting

The first piece holds chapter fixed effects constant. Within a given chapter’s territory, do statistical units with denser 1900 membership generate more post-founding patenting? The specification is

$$\ln(1 + \text{Pat}^{\text{post}})_i = \beta \ln(1 + \text{Mem}^{1900})_i + \gamma \ln(\text{Pop}_{1871})_i + \alpha_c^{\text{FE}} + \varepsilon_i, \quad (3)$$

where i is a statistical unit, c is the chapter covering i , and the chapter fixed effects α_c^{FE} absorb all chapter-level confounders — founding date, leadership, industrial mix, distance to

6. Mechanism: Amplifying Existing Capacity

the headquarters. The sample is the 357 statistical units in the canonical 22-switcher sample with non-missing members and post-founding patents.

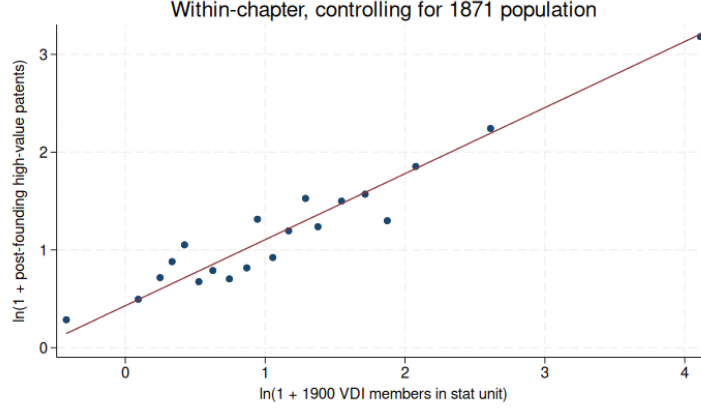


Figure 6: Within-chapter binscatter of post-founding patent intensity on 1900 member density. Twenty bins, residualised on $\ln(\text{Pop}_{1871})$ within chapter.

Figure 6 plots the binscatter. The relationship is linear in logs across the support and not outlier-driven. The regression coefficients reinforce the visual: without fixed effects, $\hat{\beta} = 0.816$; adding $\ln(\text{Pop}_{1871})$, $\hat{\beta} = 0.750$; adding chapter fixed effects, $\hat{\beta} = 0.675$ (SE 0.072). A one-percent increase in 1900 member density within a chapter’s territory is associated with a 0.68-percent increase in post-founding patenting, after every chapter-level confounder has been absorbed. The chapter is not just a marker for an industrial region; within the region, member density predicts patent intensity.

6.2. Chapters reinforce, they do not redirect

The second piece aggregates each chapter’s patent activity by sector and asks whether post-founding sectoral composition tracks pre-founding sectoral composition. For each chapter c and patent bucket b , define s_{cb}^{pre} as the chapter’s share of patents in bucket b over the ten pre-founding years $[g_c - 10, g_c - 1]$, and s_{cb}^{post} as the analogous share over the ten post-founding years $[g_c + 1, g_c + 10]$. We estimate

$$s_{cb}^{\text{post}} = \alpha_b + \beta_b s_{cb}^{\text{pre}} + \varepsilon_{cb} \quad (4)$$

bucket by bucket. A slope $\beta_b = 1$ means chapters reinforced existing specialty: a chapter that was ten percentage points heavier in bucket b pre-founding remained ten percentage

6. Mechanism: Amplifying Existing Capacity

points heavier post-founding. A slope $\beta_b = 0$ means chapters redirected: the post-founding distribution is unrelated to the pre-founding one. We restrict to the ten switching chapters with at least 20 high-value patents in both the $[g_c - 10, g_c - 1]$ and $[g_c + 1, g_c + 10]$ windows, so within-bucket shares are estimated on a non-trivial base. Table 5 reports the result on four buckets: electrical engineering and instruments, the mechanical-engineering core, the pre-industrial placebo, and a residual “Other” bucket pooling transport, food processing, and remaining sectors not covered by a 1925 technical committee.

Table 5: Post-share regressed on pre-share, by bucket: $\text{Share}_b^{\text{post}} = \alpha + \beta \text{Share}_b^{\text{pre}} + \varepsilon$. Sample: switching chapters with ≥ 20 patents in both windows $[g-10, g-1]$ and $[g+1, g+10]$; 10 chapters retained. Shares in %; robust standard errors in parentheses. The “Other (residual)” bucket pools transport, food processing, and other sectors not covered by a 1925 technical committee.

	Electrical & Instr.	VDI Core (mech. eng.)	Pre-Industrial (placebo)	Other (residual)
β (pre-share)	1.30^{***} (0.34)	0.62 [*] (0.31)	0.00 (0.17)	0.98 ^{***} (0.24)
Constant	2.71	6.11	3.18	2.29
N chapters	10	10	10	10
R^2	0.68	0.32	0.00	0.59
Effect	<i>Amplify</i>	<i>Converge</i>	<i>No signal</i>	<i>Persist</i>

The bucket-by-bucket pattern is informative. The electrical and instruments bucket shows a slope above unity ($\hat{\beta} \approx 1.30$, $R^2 = 0.68$): chapters with pre-founding strength in electrical engineering became *more* concentrated in electrical engineering post-founding — amplification of pre-existing specialty rather than redirection. The mechanical-engineering core shows a partial slope strictly between zero and one ($\hat{\beta} \approx 0.62$, marginally significant), consistent with convergence toward a common cross-chapter distribution as the Society’s standards committees spread common practice. The residual bucket shows near-full persistence ($\hat{\beta} \approx 0.98$). And the pre-industrial placebo shows a slope of essentially zero: the pre-share of pre-industrial patents carries no information for the post-share. The placebo passes within the mechanism table.

Taken as a whole, the bucket-by-bucket slopes are consistent with amplification: post-founding composition follows pre-existing specialty where the Society was organised, and shows no relationship where it was not.

6.3. Shop culture, not school culture

The third piece operates at the engineer level. Gispén’s social history of the late-19th-century German engineering profession (Gispén, 1989) distinguishes a *shop culture* — owner-engineers and senior practitioners trained on the factory floor — from a *school culture* — credentialed graduates of the technical universities. Gispén documents that chapter leadership was dominated by the shop culture; the school culture was concentrated in academic and government positions. The implication for our setting is testable: if the chapter mechanism operated through the carriers of shop-culture knowledge, the patenting response should be concentrated among owner-practitioners rather than among Dipl.-Ing. graduates.

We test the prediction with an individual linear probability model on the merged membership–patent registry. The sample is 26,869 individual members from the 1900 and 1925 directories matched against the Streb–Baten patent corpus (Section 2.2). The dependent variable is an indicator for whether the member appears as a named patent inventor in the corpus; regressors are professional-title dummies and indicators for owner-practitioner status and independent-engineer status, with the baseline category being salaried employee engineers without a Dipl.-Ing. degree. Chapter-territory fixed effects absorb regional patenting differences.

Table 6: Individual linear probability model: who patents in the engineering directory? Sample: 26,869 matched individual records from the 1900 and 1925 directories. Baseline category: salaried employee engineers without a Dipl.-Ing. degree. Chapter-territory fixed effects throughout.

	$\Delta \text{Pr}(\text{patent})$
Firm owner	+3.8pp ^{***}
Independent	+2.3pp ^{***}
Dipl.-Ing. degree	−0.9pp ^{***}
Dr. title	+1.8pp ^{***}
Seniority (years)	+0.1pp
BV territory FE	✓
<i>N</i>	26,869

The results are reported in Table 6. Owner-practitioners are 3.8 percentage points more likely to appear as patent inventors than salaried employee engineers; independent engineers are 2.3 percentage points more likely; members holding the Dipl.-Ing. title are 0.9 percentage points *less* likely. The sign flip between owners and Dipl.-Ing. graduates is exactly the prediction Gispén’s reading would generate; the empirical pattern matches it directly. A complementary cross-sectional finding, reported in the appendix, shows that the share of high-value patents

7. The Long Run: A Manufacturing Premium

filed by individual inventors declines with engineer density — consistent with organised engineering networks shifting patenting from individual inventors to corporate research, the broader structural transition of the second industrial revolution.

Each of the three pieces of evidence rules out a different alternative reading: a flat within-chapter slope would point to the chapter as a regional marker; a uniform composition effect would point to a creation mechanism; and a strong Dipl.-Ing. coefficient would point to academic spillovers. None is borne out in the data.

7. The Long Run: A Manufacturing Premium

The patent-level estimates establish a causal effect within the panel window 1877–1918. Whether the effect translated into long-run economic outcomes is a harder question, for which we offer only suggestive evidence. The long-run output panel (Peters, 2022) covers West Germany only. For identification we use a spatial donut-hole specification, mirroring the donut robustness check in Section 5.2.

We restrict the West-German modern-county sample to counties within 100 km of any pre-1918 chapter headquarters and define a donut treatment indicator T_k that equals one if county k lies 10–50 km from the nearest chapter HQ (the treat ring) and zero if it lies 50–100 km away (the control ring). Host cities (<10 km) are dropped to remove the mechanical advantage of being the chapter’s home city; counties more than 100 km away are dropped as off-support. The specification is

$$\ln Y_k = \alpha + \beta T_k + \gamma \ln(\text{Pop}_{1871,k}) + \delta_s + \varepsilon_k, \quad (5)$$

where δ_s are state fixed effects and standard errors are clustered at the nearest chapter HQ. The sample contains $N \approx 295$ –304 counties depending on outcome availability, with 34 chapter clusters.

Table 7 reports the result. The manufacturing-GDP premium in the chapter-proximate ring is sizeable and stable across the 1935–1966 window: roughly 27 log points in 1935, 26 in 1961, and 24 in 1966 — a premium of 27–31 percent across three generations and through both world wars. The corresponding GDP-per-worker premium starts at 12 log points in 1935 and fades to roughly 4 log points by 2019: the same regional advantage, attenuating gradually as broader economic convergence takes hold. Two features of the pattern align with the patent-level mechanism. The premium is much stronger on manufacturing than

8. Discussion

Table 7: Long-run donut-hole regressions. Each column is one cross-sectional regression of the long-run outcome on a donut treatment indicator (1 if county lies 10–50 km from the nearest chapter HQ, 0 if 50–100 km), with state fixed effects and $\ln(\text{Pop}_{1871})$ throughout. Host cities (<10 km) and far counties (>100 km) excluded. Sample: West-German modern counties. Standard errors clustered at the nearest chapter HQ (34 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	1935	1961	1966	1980	2019
<i>Panel A: $\ln(\text{Manufacturing GDP})$ — Peters (2022)</i>					
Donut treatment $\mathbf{1}\{10\text{--}50\text{km}\}$	0.269*** (0.093)	0.264*** (0.079)	0.241*** (0.079)	—	—
<i>Panel B: $\ln(\text{GDP per worker})$ — Peters (2022) + Destatis 2019</i>					
Donut treatment $\mathbf{1}\{10\text{--}50\text{km}\}$	0.121*** (0.036)	0.088*** (0.021)	0.069*** (0.024)	0.034 (0.033)	0.038* (0.021)
Controls: $\ln(\text{pop}_{1871})$, state FE. SE clustered at nearest chapter HQ. $N \approx 295\text{--}304$.					

on aggregate output — a four-to-sevenfold gap throughout the panel — which is where the patent-level effect operated. And it survives both world wars, suggesting persistent regional human capital and organisational know-how rather than persistent physical infrastructure. We do not claim causal identification for the long-run estimates: the donut-hole assignment is spatial proximity, not the staggered timing variation we use in the panel window, and we cannot rule out that pre-1877 conditions correlated with chapter siting also produced long-run outcomes. We report the result as evidence consistent with the patent-level mechanism, not as a separate causal estimate.

8. Discussion

The headline ATT of approximately four percent per year is large but not implausibly so. Over a 15-year post-treatment window, compounding the event-study coefficients yields a cumulative effect of roughly 60 percent of the sample-mean patent stock per statistical unit. The cohort with the longest observed post-period — chapters founded in the early 1880s — shows event-study coefficients near 10 percent by the end of the window, consistent with the linear extrapolation of the average build-up. For context, Donges, Meier, and Silva (2023) find effects of comparable order on related innovation outcomes from Napoleonic-era legal reforms; Cinnirella, Hornung, and Koschnick (2025) document large cross-sectional differences in patenting tied to economic-society membership in 18th- and early-19th-century Germany.

8. Discussion

The magnitudes we estimate sit within the range that the literature on knowledge-organisation institutions in industrial-era Europe has documented.

The central identification concern is reverse causality in founding timing: chapters may have been founded where patenting was about to accelerate. Three features of the evidence bound this concern. Switching territories were higher-patenting regions long before founding, but this level selection is absorbed by unit fixed effects; identification comes from timing, and the fifteen pre-period coefficients are flat. The hazard model finds no robust relationship between pre-period patent growth and founding timing, and the pre-period stock coefficient turns negative once 1871 population is conditioned on. And the post-treatment response builds gradually from approximately zero at founding rather than jumping — anticipatory founding would instead produce rising pre-trends and an immediate post-founding jump. What the design cannot exclude is founding in response to expectations orthogonal to observable pre-trends; with 22 switching events, randomisation inference has limited resolution, and we present the timing variation as quasi-experimental rather than as-good-as-random.

The patent-level evidence identifies a causal effect of chapter founding on high-value patenting within the panel window. It is silent on three further questions of substantive interest, each a natural extension. We do not identify the effect of the national society itself: the staggered rollout identifies the local effect of chapter founding given an existing national organisation, not the effect of the organisation’s existence in the first place. We do not identify the effect of chapter founding on innovation that fell outside the high-value patent threshold; routine technical improvements and process innovations that did not survive the ten-year renewal hurdle are not in our outcome. And we do not identify welfare consequences — patents are an intermediate output, not a welfare measure, and the long-run manufacturing results are descriptive rather than design-based.

The engineer-level analysis rests on a deliberately conservative record linkage. Of 26,869 unique directory members, 644 are linked to a high-value patent by name and residence. The low rate reflects three forces that should be kept apart. Directories list surnames with abbreviated first names, so the matching routine is tuned to avoid false positives at the cost of missed true links. High-value patenting was genuinely rare at the individual level: the outcome is a patent renewed for at least ten years, not patenting per se. Most consequentially, salaried engineers’ inventions were routinely filed under the employer’s name — the assignment question over which the Society itself lobbied in 1891 — so person-filed patents capture only the subset of inventive output its author had the right to file. The individual-level coefficients

9. Conclusion

in Section 6 should therefore be read as describing the personal-filing margin: the owner premium and the negative Dipl.-Ing. coefficient are consistent with the shop-culture account of Gispén (1989), but both are partly generated by who could file in their own name rather than by who invented. Attenuation from missed links pushes in the same direction, and we treat the engineer-level results as descriptive.

The long-run results carry caveats beyond the West-only coverage. The premium is already visible in the 1933–1939 cross-sections, which predate division and resettlement; that part cannot be produced by post-war reallocation. The post-war persistence, however, is measured on a landscape remade between 1944 and 1950: roughly eight million expellees — close to a fifth of the West German population — were resettled unevenly across counties, and firms and skilled workers relocated westward out of Berlin and the lost eastern territories. Peters (2022) shows that these expellee inflows themselves raised manufacturing employment in receiving regions. If placement or relocation correlated with distance to pre-1918 chapter headquarters, the post-war ring premium is contaminated in a direction we cannot sign; if not, the survival of the premium through a demographic shock of this size strengthens the reading of persistent regional capability. Cross-sectional variation cannot separate the two, which is why the long-run evidence remains descriptive.

9. Conclusion

The German Engineering Society’s staggered chapter rollout between 1877 and 1918 provides within-country, within-period variation in the formal coordination of practical engineering knowledge. A Callaway–Sant’Anna difference-in-differences design on the Streb–Baten corpus identifies a chapter-founding effect of approximately 4.2 percent per year. The pre-industrial-and-artisanal placebo, defined *ex ante* from the 1925 committee structure, returns a coefficient indistinguishable from zero. The result holds under alternative heterogeneity-robust estimators, sample restrictions, and inference adjustments; the donut-hole specification rules out a host-city mechanical effect; and the engineering-school placebo rules out an educational-spillover alternative. Within-design evidence indicates that chapters amplified pre-existing engineering activity rather than introducing new technology to passive regions: within chapter territory, member density predicts post-founding patenting; across chapters, sectoral patent composition tracks the host region’s pre-treatment specialty; and at the engineer level, the patenters are practical owner-engineers rather than credentialed Dipl.-Ing. graduates. A long-run chapter-proximate cross-sectional specification documents a sizeable manufacturing-GDP

References

premium for modern counties near pre-1918 chapter headquarters through the mid-twentieth century, fading on aggregate output by the present.

The broader read of the paper is that innovation capacity in industrial-era Germany was not produced by universities alone, not by formal patent institutions alone, and not by the supply of engineers alone, but by an organisational layer that coordinated practical knowledge across the divide between the academy and the factory floor. This conclusion does not displace the literature on universities (Dittmar and Meisenzahl, 2022; Squicciarini and Voigtländer, 2015), on patent rights (Moser, 2005), or on legal institutions (Donges, Meier, and Silva, 2023); what it adds is a measurement of a coordination layer that previous work has tended to leave implicit. The professional organisation of practical knowledge emerges as a distinct channel, complementary to the more formal inputs.

References

- Aghion, P. and P. Howitt (1992): A Model of Growth Through Creative Destruction. *Econometrica*, 60(2): 323–351.
- Arkhangelsky, D., S. Athey, D. A. Hirshberg, G. W. Imbens, and S. Wager (2021): Synthetic Difference-in-Differences. *American Economic Review*, 111(12): 4088–4118.
- BBSR (2019): INKAR – Indikatoren und Karten zur Raum- und Stadtentwicklung. Federal Office for Building and Regional Planning, Bonn. Online database.
- Borusyak, K., X. Jaravel, and J. Spiess (2024): Revisiting event-study designs: robust and efficient estimation. *Review of Economic Studies*.
- Bry, G. (1960): Wages in Germany, 1871–1945. Princeton University Press for the NBER.
- Callaway, B. and P. H. C. Sant’Anna (2021): Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2): 200–230.
- Cederman, L.-E., Y. Pengl, and L. Girardin (2025): RShapes: A spatial dataset of historical railway networks. Working Paper, ETH Zürich.
- Chaisemartin, C. de and X. D’Haultfœuille (2024): Difference-in-differences estimators of intertemporal treatment effects. *Review of Economics and Statistics*.

References

- Cinnirella, F., E. Hornung, and J. Koschnick (2025): Flow of ideas: Economic societies and the rise of useful knowledge. *The Economic Journal*, 135(669): 1496–1535.
- Cinnirella, F. and J. Streb (2017): The role of human capital and innovation in economic development: evidence from post-Malthusian Prussia. *Journal of Economic Growth*, 22: 193–227.
- Conley, T. G. (1999): GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1): 1–45.
- Dittmar, J. and R. R. Meisenzahl (2022): The Research University, Invention, and Industry: Evidence from German History. Federal Reserve Board Working Paper.
- Donges, A., J.-M. Meier, and R. C. Silva (2023): The Impact of Institutions on Innovation. *Management Science*, 69(4): 1951–1974.
- Gispen, K. (1989): New profession, old order: engineers and German society, 1815–1914. Cambridge: Cambridge University Press.
- Goodman-Bacon, A. (2021): Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2): 254–277.
- Hanlon, W. W. (2025): The Rise of the Engineer: Inventing the Professional Inventor During the Industrial Revolution. *The Economic Journal*, 135(670): 1749–1781.
- Kelly, M., J. Mokyr, and C. Ó Gráda (2023): The mechanics of the Industrial Revolution. *Journal of Political Economy*, 131(1): 59–94.
- Maloney, W. F. and F. Valencia Caicedo (2022): Engineering Growth. *Journal of the European Economic Association*, 20(4): 1554–1594.
- Mokyr, J. (2002): *The Gifts of Athena: Historical Origins of the Knowledge Economy*. Princeton, NJ: Princeton University Press.
- Moser, P. (2005): How Do Patent Laws Influence Innovation? Evidence from Nineteenth-Century World’s Fairs. *American Economic Review*, 95(4): 1214–1236.
- Peters, M. (2022): Historical Regional GDP Data for Germany, 1933–2019. Dataset, assembled from *Statistisches Bundesamt*, *Statistisches Reichsamt*, and *INKAR*.

References

- Romer, P. M. (1990): Endogenous Technological Change. *Journal of Political Economy*, 98(5): S71–S102.
- Rösel, F. (2023): GPOP: A historical municipality-level population panel for Germany, 1871–1950. Dataset.
- Squicciarini, M. P. and N. Voigtländer (2015): Human Capital and Industrialization: Evidence from the Age of Enlightenment. *The Quarterly Journal of Economics*, 130(4): 1825–1883.
- Streb, J., J. Baten, and S. Yin (2006): Technological and geographical knowledge spillover in the German Empire 1877–1918. *The Economic History Review*, 59(2): 347–373.
- Sun, L. and S. Abraham (2021): Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2): 175–199.
- Wasi, N. and A. Flaaen (2015): Record linkage using Stata: Preprocessing, linking, and reviewing utilities. *The Stata Journal*, 15(3): 672–697.

A. Data Construction

Membership directory transcription. The 1900 and 1925 VDI yearbooks provide complete membership rosters with names, professional titles, employers, and city of residence. We hand-transcribe both: 15,513 individual records from 1900 and 27,176 from 1925. Transcription accuracy is validated against the yearbooks’ own reported chapter-level totals; for the 1925 yearbook, transcribed counts match the published totals within 0.3 percent at the chapter level. The professional-title field is standardised to twelve categories (Dipl.-Ing., Dr.-Ing., engineer, owner-practitioner, independent engineer, employed engineer, civil servant, military, academic, student, other, missing) via a hand-coded crosswalk; the employer field is coded to firm where identifiable.

Patent–membership record linkage. The individual linear probability model in Section 6.3 requires linking the membership roster to the Streb–Baten patent corpus by inventor name. We use the probabilistic record-linkage routine of Wasi and Flaaen (2015), blocked on surname Soundex and matched on the full surname-forename-city string. Match quality is assessed manually on a random sample of 500 high-confidence matches; the false-positive rate on this subsample is approximately three percent. The pipeline yields 644 confirmed matches. The low match rate reflects three constraints: surname homonymy in 19th-century German nomenclature, transcription noise in the patent corpus’s inventor-name field, and the high renewal hurdle that restricts the patent universe to a small fraction of all filings.

Geography crosswalks. The 1900 statistical-unit shapefile is constructed from MPIDR historical-boundary files supplemented with hand-corrected boundaries where the 1871 and 1919 territorial changes affected unit boundaries. Chapter territories are taken from the 1925 yearbook’s regional map and assigned to statistical units via majority-area overlay. For robustness, a parallel pipeline constructs an independent 1882 county shapefile from the Donges, Meier, and Silva (2023) replication geography and re-runs the headline; estimates on the Donges geography are within sampling error of the 1900 statistical-unit headline.

Engineering-school and technical-university locations. The 29 state engineering school founding dates and locations are taken from Gispén (1989) and cross-checked against the *Statistik des Deutschen Reichs*. The 11 technical-university locations and founding upgrades to university status are from the same sources. The engineering-school placebo in Section 5.3

B. Identification and Estimator Detail

uses these 29 founding events; the technical-university subsample is used only descriptively, since the upgrades to university status are too synchronous (1899–1900) to support staggered identification.

B. Identification and Estimator Detail

TWFE benchmark and the Goodman-Bacon decomposition. The two-way fixed effects estimator on the canonical sample returns $\hat{\beta}^{\text{TWFE}} = 0.014$ (SE 0.008), roughly a factor of three below the doubly-robust Callaway–Sant’Anna estimate. Goodman-Bacon (2021) shows that TWFE on a staggered panel is a weighted average of all possible 2×2 DiD comparisons, including those in which already-treated cohorts serve as controls; when treatment effects build up these enter with negative weight. The decomposition on our sample places 99.8 percent of TWFE weight on legitimate timing-group comparisons: we drop the 25 always-treated chapters at the data-build stage, so the forbidden-comparison block carries weight 0.002. The TWFE–CS-DiD gap on this sample therefore does not reflect negative weights; it reflects the absence of doubly-robust inverse-propensity reweighting in TWFE, which pools heterogeneous treatment dynamics across cohorts without rebalancing on covariates. The pattern is general: in samples constructed to exclude always-treated units, the relevant attenuation channel is reweighting, not forbidden comparisons.

Alternative heterogeneity-robust estimators. The Borusyak, Jaravel, and Spiess (2024) imputation estimator returns a simple ATT of 0.043 (SE 0.004, $p < 0.01$), within sampling error of the headline and with a flat pre-period profile. The Sun and Abraham (2021) interaction-weighted estimator returns a smaller point estimate of 0.020 (SE 0.004) accompanied by a pre-period coefficient of 0.013 that is itself significant at the one-percent level; the pre-trend violation is estimator-specific — CS-DiD, dCDH, and BJS all return flat pre-trends on the same sample — and we therefore read it as an SA-internal sensitivity rather than as identification failure. The Arkhangelsky et al. (2021) synthetic DiD estimator trims chapters whose pre-period trajectory cannot be reconstructed from the available donor pool and returns 0.027 (SE 0.019) on the matched comparison; the lower point estimate reflects the smaller estimation sample, not a different identification.

C. Sample Variants and Inference Robustness

de Chaisemartin–D’Haultfœuille switchers-vs-stayers. The Chaisemartin and D’Haultfœuille (2024) estimator rests on a different identifying assumption from Callaway–Sant’Anna: it compares treatment-status switchers to stayers at each calendar year, requires no functional-form assumption on the propensity score, and does not reweight. On the canonical sample the dCDH simple ATT is 0.048 (SE 0.005, $p < 0.01$), with a pre-period coefficient of -0.0003 and a joint placebo test that does not reject (Table 8).

Table 8: dCDH switchers-vs-stayers estimator on the canonical sample.

	$\ln(1 + \text{patents})$
Pre-treatment avg	-0.0003 (0.0079)
Post-treatment avg	0.0484^{***} (0.0054)
Simple ATT	0.0484^{***} (0.0054)
N observations	20,706
N stat units	493
N cohorts	16
Event-time window	± 15
Sample period	1877–1918
Aggregation	Annual
Controls	Rail, Pop $\times$ year

C. Sample Variants and Inference Robustness

Founding-year timing of the 22 switching chapters. The identification map in the body (Figure 3) shows which territories enter the treated set; the timing dimension is reproduced as a conventional Gantt chart in Figure 7. The earliest switchers (Schleswig-Holstein and East Prussia, both 1880; Karlsruhe and Leipzig, both 1882) contribute the longest post-treatment windows. The latest, Mosel (1912), contributes only six post-treatment years and drives the lower bound of the leave-one-chapter-out range.

Excluding eastern territories. The Imperial frontier comprised East Prussia, West Prussia, Posen, and parts of Silesia — predominantly agrarian, with substantially lower urbanisation, higher-education density, and railway coverage than the western Empire. Nine of the 22

C. Sample Variants and Inference Robustness

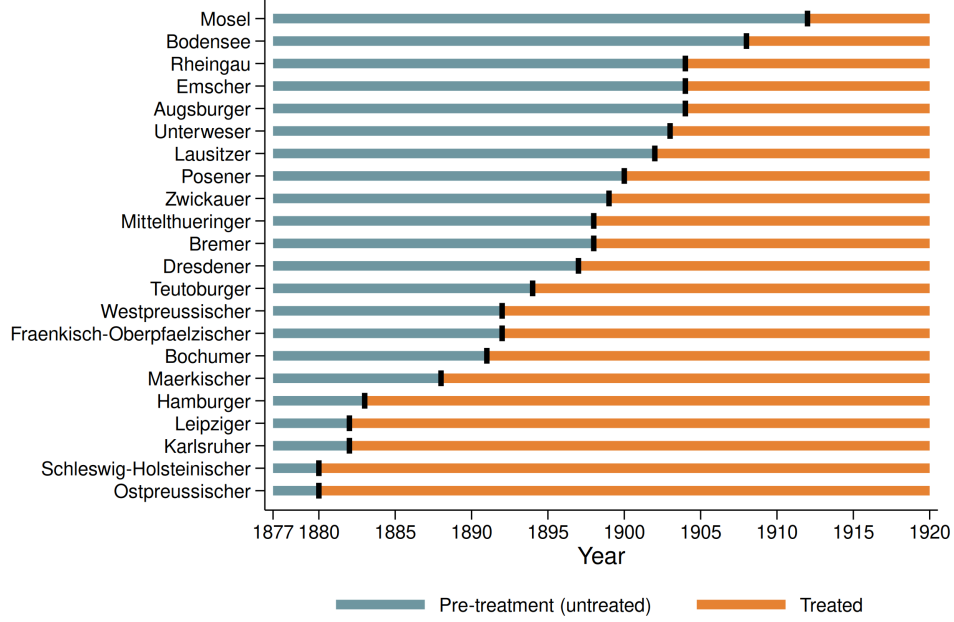


Figure 7: Staggered founding of the 22 switching chapters, 1877–1918.

switching chapters served eastern regions. Re-estimating the CS-DiD on the 13 western cohorts retained yields a simple ATT of 0.049 (SE 0.018), marginally above the full-sample headline. The argument for the exclusion is substantive: the eastern chapters served regions whose industrial composition and engineering labour market differed structurally from those for which the chapter mechanism is most likely to operate. Including or excluding them does not move the headline.

Subsample and specification variants. The pre-WWI subsample restriction (1877–1913) drops the wartime patent-office years and returns an estimate numerically identical to the full sample. The balanced-panel restriction to chapters with at least ten pre-treatment years drops three short-pre-period switchers and returns 0.070 (SE 0.016), above the full-sample headline because longer-window cohorts dominate the retained sample. Adding contemporaneous (annual) population as a control is a bad-control test, since post-treatment population is a downstream margin through which the chapter effect may propagate; the coefficient attenuates modestly to 0.043 (SE 0.018). PPML on raw counts returns 0.140 (SE 0.081, $p < 0.10$), consistent with a roughly fifteen-percent effect on the count outcome.

D. Mechanism: Supplementary Evidence

Leave-one-chapter-out and permutation inference. We re-estimate the headline 22 times, dropping each switching chapter in turn. Estimates range from 0.015 to 0.047, with 20 of the 22 drops significant at the five-percent level. The lower bound corresponds to dropping Mosel (1912, six post-treatment years); no other single chapter drives the result. A separate permutation test reassigns founding years across the 22 switching chapters 200 times; the empirical headline of 0.042 corresponds to a two-sided p -value of 0.12 (one-sided ≈ 0.06).

Selection on pre-period trends. A complementary check asks whether chapters were systematically founded in places with positive pre-period patent trends. We estimate a discrete-time hazard model in which the dependent variable is an indicator for chapter founding in year t and regressors include pre-period patent stocks, pre-period patent growth, 1871 population, and railway connectivity (Table 9). The coefficient on pre-period patent growth is positive but small and not robust; the coefficient on pre-period stocks turns negative once 1871 population is included. The pattern is inconsistent with selection on positive pre-trends.

Table 9: Discrete-time hazard: predictors of chapter founding.

	LPM hazard		Cross-section
	(1)	(2)	(3)
	$P(\text{found}_t)$	$P(\text{found}_t)$	founding year
$\ln(1 + \text{patents}_{t-1})$	0.026** (0.012)		
$\ln(1 + 3\text{yr cumul.})$		0.024** (0.009)	
$\ln(1 + \text{patents } 1877\text{-}79)$			-3.145** (1.342)
Decade FE		✓	
Sample	risk set	risk set	N=22
Cluster	chapter	chapter	robust
Obs (N)	399	355	22

D. Mechanism: Supplementary Evidence

The chapter effect scales with local member density. The headline ATT averages across all statistical units covered by a switching chapter, regardless of how many members the

E. Long-Run Robustness

chapter recruited within a given unit. Re-estimating the CS-DiD with the treatment indicator interacted with $\ln(1 + \text{members}_{1900})$ returns a positive and statistically significant interaction: doubling local 1900 member density raises the chapter effect by roughly six percentage points on a base of approximately four percent per year. In units where the chapter recruited no members, the estimated effect is statistically indistinguishable from zero. The pattern reinforces the within-chapter binscatter in the body.

Chapter founding broadens patenting. A separate outcome counts the number of distinct patent classes active in a statistical unit in a given year. Chapter founding raises this count by approximately 3.5 percent per year, consistent with the chapter mechanism broadening the technological frontier rather than merely raising filings within already-active classes. A regression on the number of patentees holding patents in two or more classes confirms a positive level effect; the share of cross-class patentees is flat, indicating that chapters grew the patenting roster as a whole rather than selectively producing generalists.

Oversupply and the credential-inflation pattern. Toward the end of the panel window the German engineering profession faced credential inflation: the rapid expansion of the technical universities after 1890 produced more Dipl.-Ing. graduates than the labour market absorbed at credentialed wages (Gispén, 1989). A seniority $\times$ Dipl.-Ing. interaction in the individual linear probability model of Section 6.3 returns a small negative coefficient (-0.003 , $p < 0.05$), consistent with credentialed graduates entering an oversupplied labour market with diminishing patenting payoffs relative to owner-practitioners.

E. Long-Run Robustness

Alternative proximity bandwidths. The body restricts the long-run sample to counties within 100 km of any pre-1918 chapter headquarters. The pattern is robust to alternative proximity thresholds. Tightening to 50 km isolates the immediate vicinity of chapter headquarters: the manufacturing-GDP premium in 1935–1966 is larger in magnitude (point estimates 0.30–0.35) but standard errors expand. Widening to 150 km halves the magnitude. Across all proximity cutoffs, the manufacturing premium exceeds the aggregate-GDP-per-worker premium by a factor of two or more and survives both world wars.

E. Long-Run Robustness

Cross-sectional persistence. The engineer-density coefficient on manufacturing GDP fades from approximately 0.30 in 1935 to 0.21 in 1966, while the coefficient on aggregate GDP per worker fades from approximately 0.14 in 1935 to 0.03 in 2019. The fading is faster on aggregate output than on manufacturing, consistent with the structural shift of West German output away from manufacturing across the second half of the twentieth century. Manufacturing persistence across both world wars and three generations of post-war restructuring is the headline long-run feature of the data.

E.1. Cross-sectional engineer-density coefficients

Table 10 reports the cross-sectional OLS regressions of long-run manufacturing GDP and GDP per worker on $\ln(1 + \text{engineers } 1900)$ referenced in Section 3. The pattern is the cross-sectional counterpart to the design-based donut-hole result in Section 7; we report it in the appendix because the cross-sectional design cannot rule out that pre-1877 conditions correlated with engineer placement also produced long-run outcomes.

Table 10: Cross-sectional OLS of long-run outcomes on $\ln(1 + \text{engineers } 1900)$ at the modern-county level. Each cell is one regression; coefficient on $\ln(1 + \text{Eng}_{1900})$ reported with SE in parentheses. State FE and $\ln(\text{Pop}_{1871})$ included throughout. Sample: West-German modern counties within 100 km of any pre-1918 chapter HQ. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Year	Manufacturing GDP	GDP per worker
1935	0.303*** (0.043)	0.139*** (0.027)
1957	0.256*** (0.030)	0.073*** (0.008)
1961	0.230*** (0.028)	0.063*** (0.008)
1966	0.206*** (0.028)	0.059*** (0.008)
1980	—	0.045*** (0.006)
2019	—	0.027*** (0.006)